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Your agent needs a sandbox, not a desert

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whoami - Samuel Colvin

- Creator of **Pydantic**
- Founder & CEO of **Pydantic Labs**

The Pydantic Stack

- Pydantic Validation: data validation library
- Pydantic AI: Agent Framework
- Pydantic Logfire: Observability - AI & Infra, Evals, Prompt management etc.

One platform. One trace.

Replace your separate infra and AI observability platforms with Pydantic Logfire.

Agents & Tools

Agents are easy to define in the abstract.

```python
env = Environment()
tools = Tools(env)
system_prompt = "Goals, constraints, and how to act"

while True:
    action = llm.run(system_prompt + env.state)
    env.state = tools.run(action)
```

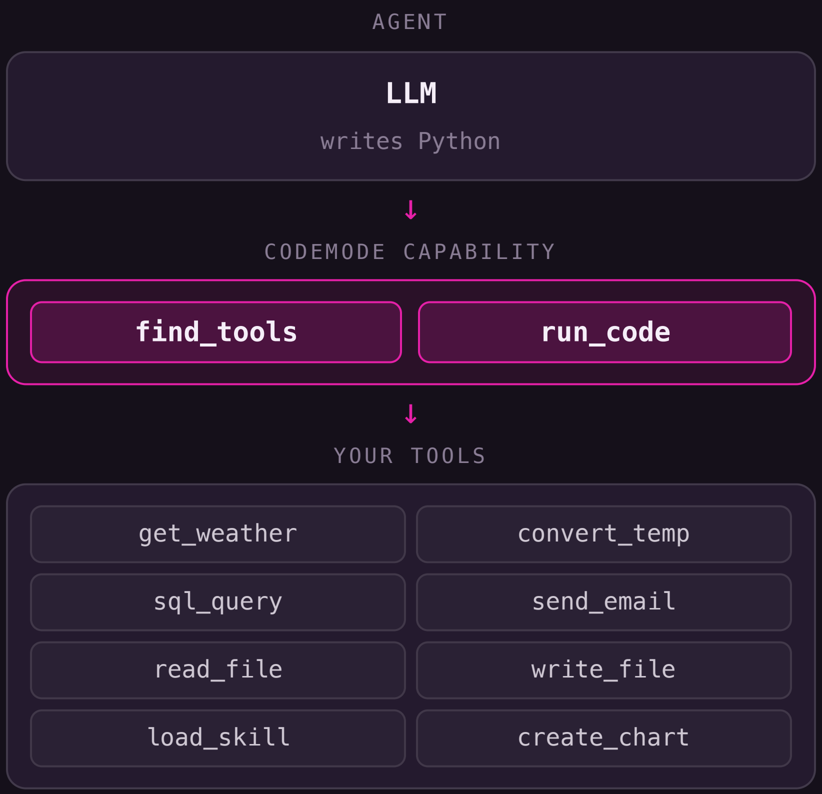
Agents are models using tools in a loop

... but how many tools?

```
{
  "get_weather",
  "search_flights",
  "book_hotel"
}
```

Enter Codemode 🥁

- aka "Programatic tool calling", CodeAct, and RLM (Recursive Language Models)
- The model writes Python (or js); the Python calls tools
- Tool composition, branching, looping - all happen in **code**, not round-trips
- Intermediate state stays in the runtime, not in the context window
- The agent only ever sees **two** tools: `run_code` and `find_tools`
- "1000 tools in 1000 tokens" is the promise
- even better "1000 tool calls with 1 LLM call"



But how do we run it?

The people who benefit most from codemode are those selling sandboxes.

Two Paradigms

Both alike in dignity.

Light tasks

- Pass tool outputs to next tool
- Calculations
- Run queries
- Render a chart

TYPICAL DURATION

`milliseconds`

Heavy tasks

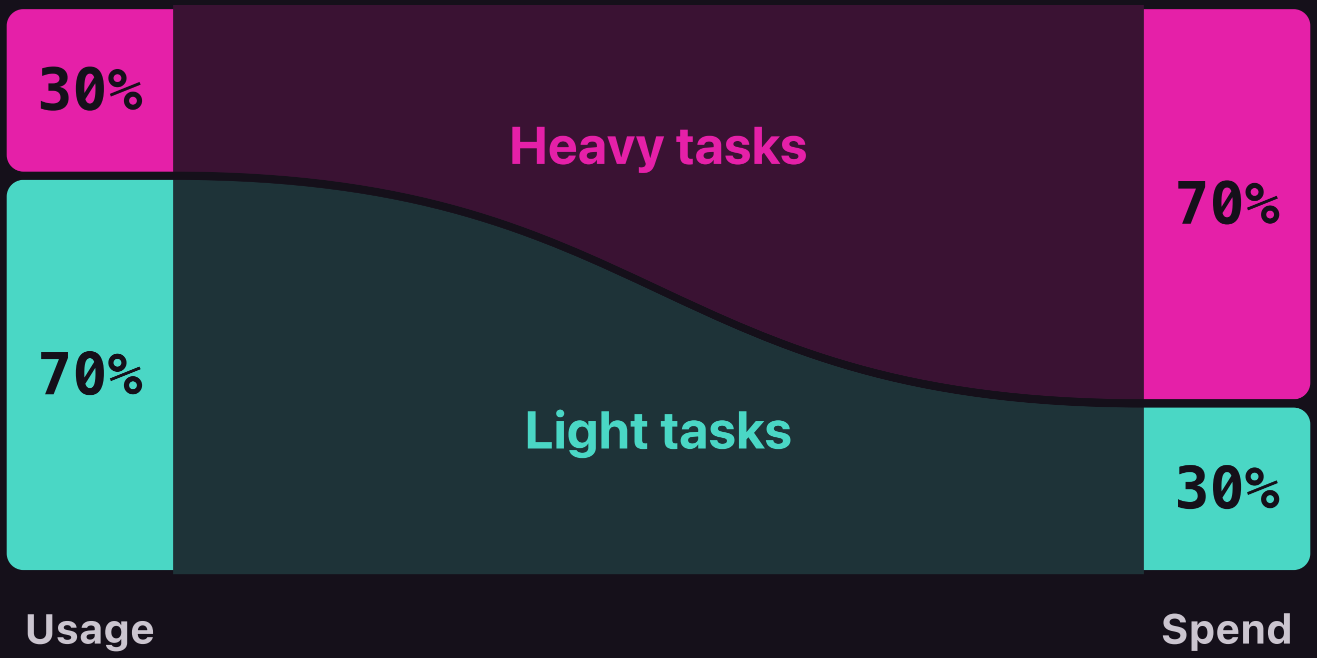
- Clone a repo
- Run a coding agent
- Heavy computation
- Compilation

TYPICAL DURATION

`minutes`

Two Paradigms

Both alike in dignity. **Usage**



Two Paradigms

Both alike in dignity. **Problems**

Light tasks

Execution environments:

- VM
- Container
- gVisor
- Firecracker

TYPICAL BOOT TIME

1–3s

Heavy tasks

Execution environments:

- VM
- Container
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TYPICAL BOOT TIME

1–3s

The sandbox vs. the desert

THE DESERT

awk sed gcc g++ clang less more
grep egrep fgrep make cmake
bash zsh sh fish curl wget
tar gzip bzip2 xz zip unzip
find xargs locate vim nano
emacs ed python3 pip pipx node
npm pnpm yarn git svn hg ssh
scp rsync cron systemd apt
apt-get dpkg yum dnf brew jq yq
tr cut sort uniq head tail

THE SANDBOX

`sql_query`

`create_chart`

`create_table`

`load_skill`

Sandbox, not a desert

- A **curated list** beats the kitchen sink
- Most agents don't need a whole Linux box - they need ~5 tools used well
- "Analyse BI data to find efficiencies" sounds broad - the actual surface is tiny
- Full sandboxes solve a problem most embedded agents don't have

What would a better sandbox for light tasks look like?

Introducing Pydantic Monty

- A **minimal Python interpreter, written in Rust**, for code written by AI
- Safe by default: no filesystem, env, or network unless you grant it
- Pass in your own functions - the model calls them as ordinary Python
- **Type-checks** code (via `ty`) before it runs
- **Snapshottable** - pause at an external call, resume later, anywhere
- Supports a small but growing subsection of the standard library - `re`, `json`, `datetime`, `Pathlib` etc.
- Monty code is run in a pool of subprocesses, wire protocol for communicating with the main process

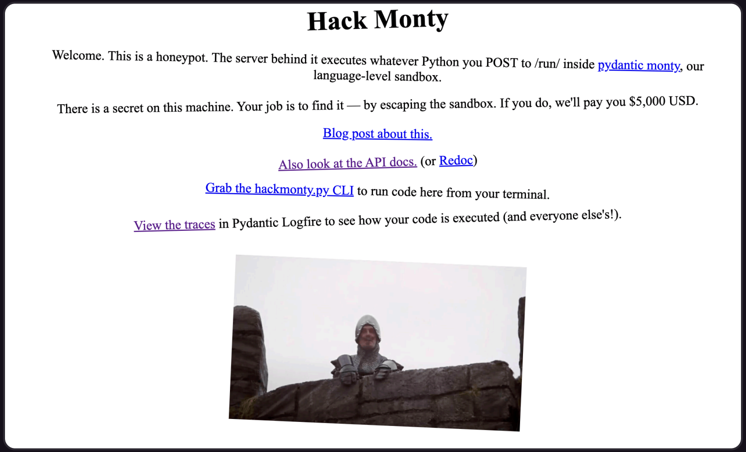
```
➤ uv add pydantic-monty
➤ npm i @pydantic/monty
➤ cargo add monty
```

The advantages

STARTUP LATENCY	5µs MONTY	vs.	~~1s~~ FULL SANDBOX
SNAPSHOT SIZE	1KB MONTY	vs.	~~1GB~~ FULL SANDBOX
COST PER RUN	$0 MONTY	vs.	~~$?~~ FULL SANDBOX

Hack Monty

- Public bounty at hackmonty.com
- Scape the sandbox, read the secret and we'll pay **$10,000**
- Round 2 launched a month ago
- No sandbox vulnerability found



Demo

🤞

Coming soon...

We also have a plan to solve agents for heavy tasks:

```
async with logfire.sandbox(provider='modal', api_key=LOGFIRE_TOKEN) as session:
    result = await session.run_code(code, external_functions={'my_local_function': ...})
```

Thank you

pydantic.dev/contact

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